Malak Al-Hanafi

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Professional Summary

AI and Mechatronics Engineer with hands-on experience building LLM-powered applications, REST API integrations, and full-stack software in Python, Node.js, and TypeScript. Combines practical engineering, AI research, and strong technical communication to build reliable, real-world solutions.

Education

Western University — London, ON
BESc, Mechatronics and AI Systems Engineering (Dual Degree) — Jun 2026

- Engineering Leadership and Innovation Certificate, Dean's Honour List

Relevant Experience

Autonomous Vehicles & Robotics Researcher — Sep 2026 – Jan 2027
King Abdullah University of Science and Technology (KAUST) — Thuwal, Saudi Arabia

- Selected for a competitive research program, collaborating with AI and cybersecurity researchers to develop intelligent systems focused on reliability, security, and real-world engineering challenges.
- Developing solutions that help connected autonomous vehicles distinguish technical failures from security threats, improving how intelligent systems evaluate trust and make reliable decisions.
- Designing an AI-powered interaction system that combines video, audio, and language-model-generated data to interpret customer behaviour and support intelligent responses in service robots.

AI Engineering Intern — May 2026 – Jul 2026
SpeedLogix Group — Oakville, ON

- Developed Node.js and REST API integrations connecting an LLM-powered voice assistant with backend services, enabling real-time communication between conversational AI and application systems.
- Collaborated with research and product teams to translate stakeholder requirements into technical implementations, aligning AI functionality with business needs and project priorities.

Machine Learning Teaching Assistant & Lab Instructor — Sep 2025 – Jan 2026
King's University College — London, ON

- Designed and delivered 12 hands-on machine learning labs (view labs) for 20+ students, translating complex technical concepts into practical, accessible exercises using Python and R.
- Collaborated with the course instructor to develop structured learning materials, assess technical work, and provide clear, constructive feedback.

Full-Stack Engineering Intern — May 2024 – Sep 2024
Smart Methods — Jeddah, Saudi Arabia

- Developed full-stack software features and REST API integrations across multiple client projects, adapting implementations to different technical requirements using Python, JavaScript, and web technologies.
- Awarded Best Engineer Intern across the internship cohort in recognition of technical contributions and delivery quality.

Selected Projects

AI-Enabled Autonomous Lawn Mower (GitHub repo) | *Jetson Nano* | *YOLOv8* | *PyTorch* | *OpenCV* | *ROS*

- Led AI and perception development for an autonomous lawn mower, integrating computer vision, navigation, and automated safety mechanisms into a ROS-based robotic system.
- Worked with a multidisciplinary engineering team to combine software and hardware components, implementing layered safety checks using cameras, LiDAR, and ultrasonic sensors.
- Contributed to a capstone project that earned an Honourable Mention at Engineering Design Day.

Autonomous Navigation for Wheelchair Users | *Python* | *OpenCV* | *Machine Learning*

- Designed an autonomous navigation system for power wheelchair users, applying computer vision and software automation to address real-world mobility challenges.

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, R, MATLAB, HTML, CSS
Software Engineering & Tools: Node.js, REST APIs, Full-Stack Development, Git/GitHub, Docker, Kubernetes, GCP, Claude Code, GitHub Copilot
AI & Machine Learning: LLM Integration, Conversational AI, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLOv8